

Chapter 17: Diseases of Fibre Crops

Cotton (Gossypium Linn.)

Leaf Spot Disease of Cotton

The pathogen causing this disease is *Cercospora gossypina* Cke. In its systematic position, it belongs to Class Deuteromycetes, Order Moniliales, Family Dematiaceae.

Distribution

The disease is found in many countries of the world such as the United States, West Indies, Egypt, China, and other cotton growing countries. In India, this disease is found sporadically in cotton growing states.

Symptoms

The disease may appear at any stage in the growth of the plant, but chiefly when plants are fully grown. The rounded or irregular spots appear on both surfaces of the leaf, and are generally 4 mm in diameter. They are at first yellowish-brown, then whitish in the centre from the withering of the tissues. The margin of the spot consists of a distinct, dark brown or a blackish rim which in young spots may be reddish in the beginning. The adjacent spots may coalesce with each other, forming large patches of the leaf to wither. In old spots, the centre cracks or breaks, forming a hole. Badly infected leaves lose their green colour, turn yellow, and ultimately wither and drop off.

An infected leaf shows numerous leaf spots on its surface, with black dots visible within the spots that represent tufts of conidiophores and conidia. The conidiophores are slender and septate, and the conidia are transversely septate and elongated.

The Pathogen

The leaf spot disease of cotton is caused by *Cercospora gossypina* Cke. The mycelium is strictly confined to the spots. The hyphae are intercellular, hyaline, and septate. After growth for a time, small stromatic masses of dark hyphae collect near the surface, chiefly on the upper side of the leaf. From these stromatic masses of hyphae, tufts of conidiophores are given out, usually through the stomata. The conidiophores are up to 150 microns long, flexuous, dark-brown, and septate. The conidia develop at the apices of conidiophores; they are hyaline, long, slender, curved bodies, narrow above and rounded below, 5 to 7 times septate, and measure 70 to 180 by 3 microns.

The perfect perithecial stage is represented by an ascomycete, *Mycosphaerella gossypina* (Atk.) Earle, which has been found in the United States and China. The perithecia are ovate, black, partly sunken in the tissue, and measure 60 to 70 by 65 to 91 microns. The asci are sub-cylindric and measure 8 to 10 by 40 to 45 microns. The ascospores are elliptic with slight constrictions at the septa, measuring 15 to 18 by 3 to 4 microns.

Nature and Recurrence of Disease

This is a soil-borne disease. The mycelium and conidia survive in the soil and infect the plants in the coming season.

Predisposing Factors

Wet soil and misty weather favour the disease incidence. Attack on the young parts of the plant usually indicates want of vigour or an unfavourable season.

Control Measures

The exotic varieties of cotton are immune to the disease. Since the disease is soil-borne, sanitation should be practiced: all plant debris should be collected and destroyed by burning it.

Root Rot of Cotton

The pathogen causing root rot of cotton is *Rhizoctonia bataticola* (Taub.) Butl., also known by the names *Rhizoctonia solani* and *Macrophomina phaseolina*. In its systematic position, it belongs to Class Deuteromycetes, Order Sphaeropsidales, Family Sphaeropsidaceae.

Distribution

This disease occurs in several countries of the world, such as Venezuela, U.S.A., India, and parts of Africa. In India, the disease has been recorded from several states such as Madhya Pradesh, Karnataka, and Tamil Nadu.

Symptoms

This fungus causes chiefly a seedling disease. Infected seedlings usually show a soft, yellow patch on the lower portion of the stem. In slightly older seedlings, when the stem is more woody, it becomes black at the ground level. On pulling out the infected plants, the entire root system may be found rotting and decaying. In bad cases, the rot is enough to cause tender plants to collapse, while older plants dry up and wither. The fungal sclerotia appear as minute black dots on the surface of woody tissue and bark.

An infected root shows rounded sclerotia, and the fungal hyphae are thick-walled. The sclerotium and hyphae, as well as the pycnidium and its conidia, are characteristic features of the pathogen.

The Pathogen

The root rot of cotton is caused by *Rhizoctonia bataticola* (Taub.) Butl. The hyphae are formed in the rotted area, at first in the cortex, and later in bad cases extending up to the pith. The fungal hyphae are found inter- and intracellularly in the host tissue. The hyphae are septate and thick. The sclerotia are produced among the hyphae on the surface of the infected plant. Each sclerotium measures 100 microns in diameter.

Nature and Recurrence of Disease

This is a soil-borne disease. The fungus can survive for many years in soil. The fungus attacks the roots of the host plant. As soon as the hyphae establish in the roots, they multiply rapidly in the cortical tissues, extend up to the pith, and ultimately enter the xylem vessels. The pathogen survives in the soil, and when the cotton crop is sown again in infested soil, the fungus infects the plants.

Control Measures

Since the pathogen survives in the soil for many years, a suitable crop rotation is of much use for controlling the disease. The sowing of resistant varieties is the best method of control, but sufficient work has not been done in this direction. As a cultural practice, the sowing dates of the crop may be so adjusted that the seedling stage is avoided when the soil temperature is high, above 35 degrees Celsius. Mixed cropping with sorghum and *Phaseolus* is helpful in reducing the soil temperature.

Wilt of Cotton

The pathogen causing wilt of cotton is *Fusarium oxysporum* f. *vasinfectum* (Atk.) Snyder and Hansen, with the synonym *F. vasinfectum* Atk. In its systematic position, it belongs to Class Deuteromycetes, Order Moniliales, Family Tuberculariaceae.

Distribution

The disease is found all over the world wherever the crop of cotton is grown. It is believed that it has started from Mexico or Central America. The disease has been reported from many other countries such as Africa, France, Russia, Italy, and the West Indies. In India, the disease has been recorded from several states, especially Maharashtra, Madhya Pradesh, Gujarat, and other parts of Central India. The disease is restricted to the black cotton soils, which are heavy clay with an alkaline reaction, but in the light alkaline and loamy soils it is unknown.

Symptoms

The fungus attacks the host when the plants are 1 to 3 weeks old. The disease becomes manifest when the plants are 5 to 6 weeks old, at which time the leaves become yellow and wither, until finally the entire plant dries up. The plants soon wilt and die. In young affected plants the basal part becomes black, whereas in mature plants examined, partial discolouration is seen. If an attacked plant is pulled up and examined, it will be found that the vascular tissues of the roots and stem are blackened because of the plugging of vessels. As a rule the affected plants succumb completely to the disease, but cases of partial infection are also not uncommon. When examined, the transverse sections of the affected stem show that the vascular bundles are seen plugged with fungal hyphae and a gummy substance.

A wilted plant shows blackened roots and wilted leaves. In transverse section, the mycelium can be seen within the vessels of the infected plant, along with the microconidia and macroconidia of the fungus.

The Pathogen

The wilt of cotton is caused by *Fusarium oxysporum* f. *vasinfectum* (Atk.) Snyder and Hansen. The fungus is found both inter- and intracellularly in the host tissue. The mycelium of the fungus plugs the xylem vessels. Hyphae run through the vessels and permeate the stem and the complete root system. The plugging of vessels interferes with the flow of water, and wilting of leaves results. The fungus secretes toxic substances, which also cause the wilting of the plant. The mycelium is hyaline, and produces macro- and microconidia. The macroconidia are 1 to 5 septate, hyaline, thin-walled, linear, and falcate, whereas the microconidia are elliptical to spherical, thin-walled, and one or two-celled. The chlamydospores are formed only in culture media.

Nature and Recurrence of Disease

This is a soil-borne as well as seed-borne disease. This fungus is a facultative parasite and can remain viable in the soil for a long time. The fungus enters the host root when the plants are 1 to 3 weeks old. The fungus multiplies and moves to the xylem vessels, and ultimately the vessels become plugged by an abundance of mycelium. The soil-borne fungal hyphae and chlamydospores play the major role in perpetuation of the disease.

Predisposing Factors

The soil temperature ranging from 20 to 30 degrees Celsius favours the disease incidence. Hot and dry periods followed by rains also favour the disease. Nitrogenous fertilizers increase the incidence. The disease is restricted to the black cotton soils, which are heavy clay with an alkaline reaction.

Control Measures

The use of resistant varieties is the only appropriate method of controlling this disease. Resistant varieties which have the desired agronomic and commercial characters have been evolved and cultivated in different states of India. In India, all the New World Cottons (*Gossypium hirsutum*, *G. barbadense*) are immune to wilt, even those which are highly susceptible to the disease in the countries from which they have been imported, for example, the United States. Wilt occurs in that country in sandy acidic soils, and when those American varieties are grown in the heavy clayey and alkaline soils of India, they acquire immunity to the disease.

As a seed treatment, the seed-borne phase can be destroyed by treating the seeds with organomercurials to some extent. Regarding exposure of soil to sun, in north India, after deep ploughing, the soil is exposed to sun in June-July to kill the pathogen. Proper manuring is also important: potash application reduces wilt incidence, whereas nitrogenous fertilizers are responsible for increasing disease incidence.

Angular Leaf-Spot, Blackarm and Boll Rot of Cotton (A Bacterial Disease)

The pathogen causing this disease is *Xanthomonas malvacearum* (E.F. Smith) Dowson. In its systematic position, it belongs to Class Schizomycetes, Order Pseudomonadales, Family Pseudomonadaceae.

Distribution and Background

This disease occurs in all cotton-growing countries of the world. The disease is responsible for heavy losses and has been supposed to be the third major disease of cotton. It was first recorded in the United States of America by Smith in 1920, where it was known as angular leaf-spot, as the spotting of the leaves was the most obvious symptom. Smith also established the connection between the leaf spot, the stem infections known as blackarm, and the attack on the bolls. The stem infections, known as blackarm, cause more serious damage than the leaf-spot.

Symptoms

On the leaves, small spots appear, angular in shape, as the primary infections are limited by the small veins. The spots are dark brown in colour, and they may coalesce to form large blotches on newly formed leaves; the attack is limited to the tissue on either side of the main veins. In wet weather, the

narrow dark-brown areas extend up to the petioles. However, in dry weather conditions, the severely attacked leaves fall off, and the plants remain barren. Elongated and discoloured areas appear on the young shoots and older branches. The stem infections are sunken and turn black, from which symptom the name blackarm has been derived; such symptoms may extend many inches in length.

Affected leaves show lesions on the leaf blade, and similar lesions can be seen on the stems and seedlings.

The bolls may be attacked at all stages of development. The young bolls may be killed and fall off, whereas the older bolls bear large dark spots which turn black. The seeds also become contaminated, and the lint is often destroyed or rendered useless.

The Pathogen

The angular leaf-spot or blackarm disease of cotton is caused by a bacterium, *Xanthomonas malvacearum* (E.F. Smith) Dowson. The bacterium is a short motile rod, provided with a single polar flagellum. It is gram-negative, and measures 1.0 to 1.2 by 0.7 to 0.9 microns.

The bacterium does not enter the vascular bundles but is confined to the parenchyma. From the leaf-stem lesions, the bacteria are disseminated by wind-blown rain onto the bolls, where spots are formed. In the infected capsules, the bacteria penetrate to the developing seed and form slimy masses around the hairs surrounding the micropyle.

Nature and Recurrence of Disease

This is a seed-borne disease. The primary infection takes place through seed-borne bacteria. The bacteria form slimy masses around the hairs surrounding the micropyle of the seed. On germination of contaminated seed, the bacteria infect the cotyledons as they emerge through the soil and produce elongated marginal lesions. Thereafter, the leaves and stems become infected.

The bacteria do not survive long in the soil. In plant debris the organism can survive for a few weeks, and sometimes serves as a means of inoculation.

Predisposing Factors

The spread of the disease depends much upon climatic conditions. In rainy weather the leaves and stems are attacked, especially when the rains are heavy and accompanied by strong winds. The pathogen is carried by wind-driven rain onto the plants. The most important factors favourable to the pathogen are high relative humidity and high air temperature.

Control Measures

Since the disease is seed-borne, the seeds should be treated with organo-mercurials prior to sowing, as a seed treatment. The seeds may also be soaked overnight in 1000 ppm of streptomycin sulphate. Since the bacteria can survive for a few weeks in plant debris, sanitation should be practiced: the affected portions should be collected and destroyed. Regarding resistant varieties, these should be evolved; in India, work is in progress in this direction with good prospects.